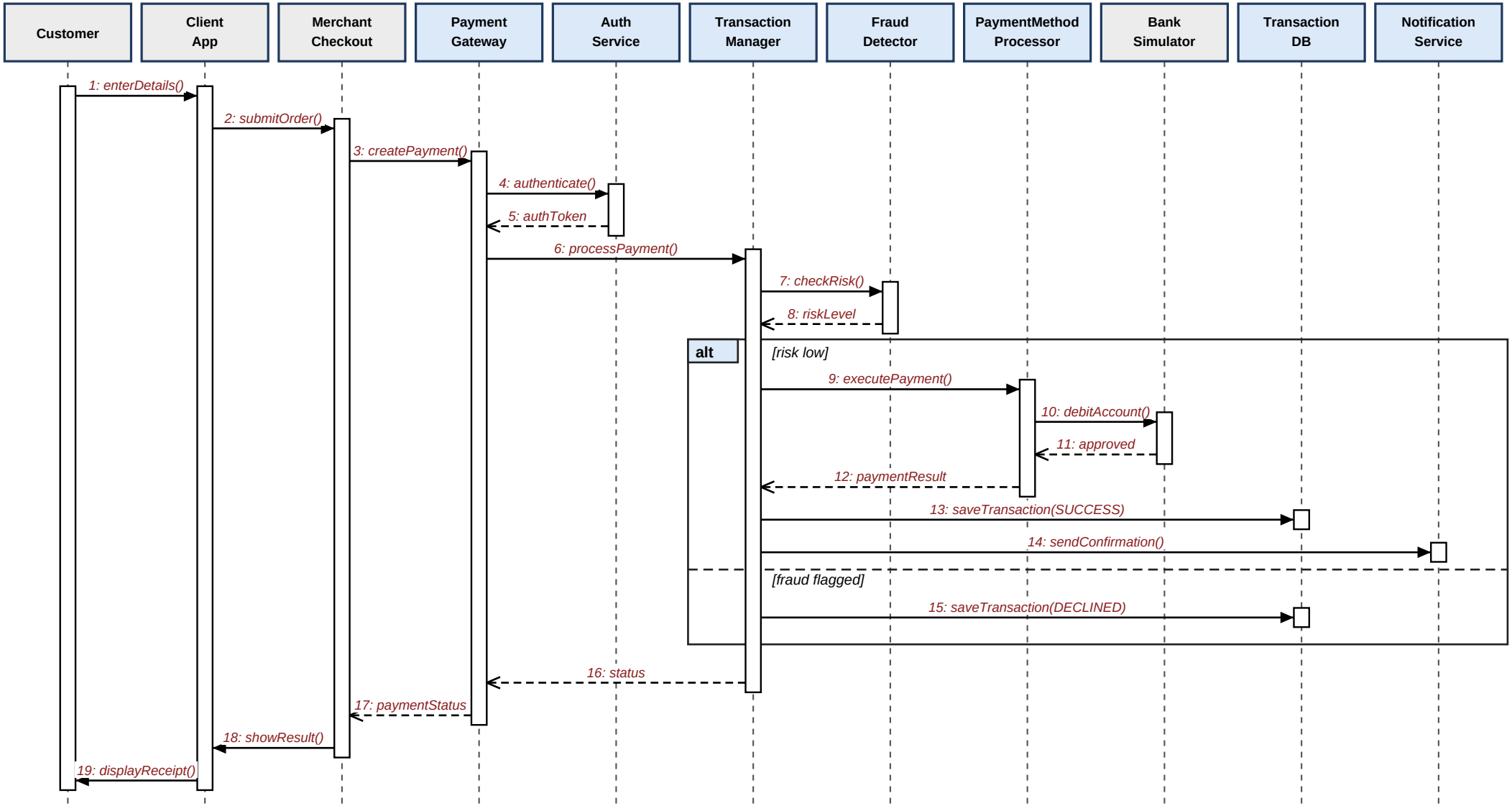
KIET Deemed to be University, Ghaziabad

Assignment: UML Sequence Diagram (Successful and Declined Payment)

Name: Ashutosh Tiwari Branch: CSE Section: B Semester: V

University Reg. No.: 202401100200098 University Roll No.: 2400290100098 Project: Digital Payment Gateway Simulation



Solid arrow = call / request Dashed arrow = return / response alt = alternative flows

Fig.: Sequence Diagram of a Payment in the Digital Payment Gateway Simulation

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Assignment: UML Sequence Diagram - Message Description

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No.	From	To	Message	Description
1	Customer	ClientApp	enterDetails()	Customer types card/UPI/net-banking details
2	ClientApp	MerchantCheckout	submitOrder()	Client app sends the order to the merchant
3	MerchantCheckout	PaymentGateway	createPayment()	Merchant asks the gateway to start a payment
4	PaymentGateway	AuthService	authenticate()	Gateway asks the authentication service to verify the user
5	AuthService	PaymentGateway	authToken (return)	Returns token after OTP/PIN is verified
6	PaymentGateway	TransactionManager	processPayment()	Gateway forwards the payment to the transaction manager
7	TransactionManager	FraudDetector	checkRisk()	Transaction manager asks for a fraud check
8	FraudDetector	TransactionManager	riskLevel (return)	Returns LOW or HIGH risk
9	TransactionManager	PaymentMethodProcessor	executePayment()	[risk low] Routes payment to the selected method handler
10	PaymentMethodProcessor	BankSimulator	debitAccount()	Handler requests debit from the bank simulator
11	BankSimulator	PaymentMethodProcessor	approved (return)	Bank approves (balance sufficient)
12	PaymentMethodProcessor	TransactionManager	paymentResult (return)	Handler returns the result
13	TransactionManager	TransactionDB	saveTransaction(SUCCESS)	[risk low] Stores transaction as SUCCESS
14	TransactionManager	NotificationService	sendConfirmation()	[risk low] Sends SMS/e-mail confirmation
15	TransactionManager	TransactionDB	saveTransaction(DECLINED)	[fraud flagged] Stores transaction as DECLINED
16	TransactionManager	PaymentGateway	status (return)	Transaction manager returns final status to gateway
17	PaymentGateway	MerchantCheckout	paymentStatus (return)	Gateway returns status to the merchant
18	MerchantCheckout	ClientApp	showResult()	Merchant shows the result
19	ClientApp	Customer	displayReceipt()	Client app shows receipt to the customer